∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = 0

∴ The condition that for each column containing a leading entry, all positions below it must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 3 = 0

∴ The condition that for each column containing a leading entry, all positions above it must contain 0 is True.

The final matrix $\begin{bmatrix} 1 & -5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is in rref. □

By defn, both Rows have pivots.

$\boxed{\therefore T: \mathbb{R}^3 \to \mathbb{R}^2 \text{ is a linear and onto transformation}}$ □

It's easy to see that the corresponding rref of $[A \ 0]$ is $\begin{bmatrix} 1 & -5 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. There is no row in the matrix of the form $[0 \ 0 \ 0 \ldots\ldots 0 \ b]$ where $b \neq 0$. By Theorem 2, the corresponding system of equations is consistent.

- By Defn, x_2 is a free variable since Column 2 has no pivots.
- By Theorem 2, the matrix equation $Ax = 0$ has infinitely many solutions.

$\boxed{\therefore T: \mathbb{R}^3 \to \mathbb{R}^2 \text{ is not a one-to-one transformation}}$ □

In Exercises 29 and 30, describe the possible echelon forms of the standard matrix for a linear transformation T. Use the notation of Example 1 in Section 1.2

29. $T: \mathbb{R}^3 \to \mathbb{R}^4$ is one-to-one.

Solution: By Theorem 10, the standard matrix of T is the unique 4x3 matrix A. By Theorem 11, since T is one-to-one, the equation $T(x) = 0$ has only the unique trivial solution. ∴ $Ax = 0$ has only the unique trivial solution.